Midterm Exam Grading Rubric

IEDA 1250: Optimization for Decision Making (Summer, 2026)

Question 1: Linear Programming – Graphical Method (10 Points)

- Correctly identifies that the feasible region is a diamond bounded by $0 \le x_1 + 2x_2 \le 6$ and $-2 \le 2x_1 - x_2 \le 4$. **(2 Points)**
- Correctly finds the four vertices of the feasible region:
$$\left(-\frac{4}{5}, \frac{2}{5}\right), \left(\frac{8}{5}, -\frac{4}{5}\right), \left(\frac{2}{5}, \frac{14}{5}\right), \left(\frac{14}{5}, \frac{8}{5}\right).$$
(4 Points)
- Correctly identifies the optimal solution and optimal objective value:
$$x_1^* = \frac{14}{5}, \qquad x_2^* = \frac{8}{5}, \qquad z^* = \frac{18}{5}.$$
(4 Points)

Question 2: Linear Programming – Complementary Slackness (20 Points)

- Correctly writes or uses the dual constraints corresponding to the primal LP, taking into account that x_3, x_4, x_5 are unrestricted in sign. **(4 Points)**
- Correctly applies complementary slackness to the primal constraints and obtains
$$2x_1 + x_2 + x_3 + x_4 = 3.$$
(4 Points)
- Correctly applies complementary slackness to the dual constraints for x_1 and x_2, and concludes that
$$x_1 = 0, \qquad x_2 = 0.$$
(5 Points)
- Correctly substitutes $x_1 = 0$ and $x_2 = 0$ into the primal constraints and forms the reduced system:
$$\begin{cases} x_3 + x_4 = 3, \\ 2x_3 + x_5 = 7, \\ x_3 - x_4 + 2x_5 = 7. \end{cases}$$
(4 Points)
- Correctly solves the reduced system and obtains the optimal primal solution:
$$x_1^* = 0, \quad x_2^* = 0, \quad x_3^* = 2, \quad x_4^* = 1, \quad x_5^* = 3.$$
the optimal objective value:
$$z^* = 6.$$
(3 Points)

Question 4: Mixed-Integer Linear Programming – Production Planning (20 Points)

- Correctly defines production, inventory, and setup decision variables, including binary setup variables y_t for each month. **(4 Points)**
- Correctly formulates the objective function with production costs, fixed setup costs, and holding costs for Months 1–5. **(4 Points)**
- Correctly writes the monthly inventory balance constraints using zero initial inventory and all monthly demands. **(5 Points)**
- Correctly imposes no ending inventory, $I_6 = 0$, and nonnegativity of production and inventory variables. **(3 Points)**
- Correctly links production and setup decisions using capacity constraints such as $0 \leq x_t \leq 80y_t$ and states $y_t \in \{0, 1\}$. **(4 Points)**

Question 5: Game Theory – Dominance and Pure Strategy Equilibrium (20 Points)

- Correctly recognizes that Player A maximizes the payoff and Player B minimizes the payoff in the zero-sum game. **(2 Points)**
- Correctly eliminates dominated rows for Player A: Rows $A3, A4, A5$ are eliminated. **(5 Points)**
- Correctly eliminates dominated columns for Player B: Columns $B3, B4, B5$ are eliminated. **(5 Points)**
- Correctly obtains the reduced game

	$B1$	$B2$
$A1$	4	1
$A2$	2	3

  (3 Points)
- Correctly checks the saddle point condition: $\max \min = 2$ and $\min \max = 3$, so no pure strategy Nash equilibrium exists. **(3 Points)**
- Correctly explains why each payoff in the reduced game is unstable by identifying a profitable unilateral deviation by Player A or Player B. **(2 Points)**

Question 6: Integer Programming – Logical Constraints (30 Points)

- Correctly defines binary project selection variables $x_i \in \{0, 1\}$ and formulates the profit-maximization objective and budget constraint. **(4 Points)**

- Correctly formulates constraints (a)–(h):

$$x_2 + x_5 \le 1, \quad \sum_{i=1}^{10} x_i = 4, \quad x_1 + x_3 + x_4 \ge 1,$$

$$x_6 \le x_7, \quad x_8 \le x_3 + x_4, \quad x_1 = x_{10},$$

$$x_9 \ge x_3 + x_4 - 1, \quad x_6 + x_7 + x_8 \ge 2x_5.$$

  (14 Points)

- Correctly models condition (i), including an auxiliary binary variable and valid linear constraints enforcing at least one of the two stated conditions. One valid formulation is

$$x_1 + x_2 + x_3 + x_4 \ge 2z_i,$$

$$x_5 + x_6 + x_7 + x_8 \le 1 + 3z_i, \qquad z_i \in \{0,1\}.$$

  (5 Points)

- Correctly models condition (j), including an auxiliary binary variable and valid linear constraints for the conditional disjunction. One valid formulation is

$$x_2 \ge x_{10} + z_j - 1, \qquad x_9 \ge x_{10} + z_j - 1,$$

$$x_3 \le 1 - x_{10} + z_j, \qquad x_4 \le 1 - x_{10} + z_j, \qquad z_j \in \{0,1\}.$$

  (5 Points)

- Uses clear notation and states all required binary restrictions for auxiliary variables. **(2 Points)**